

Assessment 3

OpenCV Image Creation, Segmentation & Composition



This assessment introduces practical OpenCV image creation, foreground segmentation, masking, transparency, and background composition.

Projects

1. Draw the OpenCV Logo Using Geometric Shapes

Recreate the supplied OpenCV logo reference using OpenCV drawing functions:

- `circle()`
- `fillPoly()`
- `putText()`
- NumPy coordinate arrays

The three colored logo pieces are constructed from geometric shapes rather than using the supplied logo as an image.

Input: No external input required.

Output: `opencv_logo_output.png`

Run: `python opencv_logo.py`

2. Separate Foreground from Background

Segment the supplied foreground object from its background and produce a foreground-only image and a transparent PNG for the next project.

Key skills: color segmentation, masking, morphological operations, connected components, foreground extraction, and transparent PNG creation.

3. Replace the Background

Use the transparent foreground generated by Project 2 and place it onto a new background using alpha blending.

Key skills: image resizing, ROI placement, alpha masks, image composition, and blending.

Assessment Flow

Project	Main Skill	Output
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1. OpenCV Logo	Geometric drawing	Generated logo
2. Foreground Separation	Segmentation & masking	Foreground + transparent PNG
3. Background Replacement	Composition & alpha blending	Final composite image

Learning Outcome: Learners practice creating images with OpenCV primitives, separating foreground objects, creating transparent masks, and composing images onto new backgrounds.